Data Analytics Capstone

Synopsis

Early Detection of Coronary Heart Disease Using Machine Learning

Vinaya Bhushan M
Walsh College
QM640: Data Analytics Capstone
Mentor: Vikas S
Fall 2025, Term 3

1. Introduction

1.1 Background and Context

Coronary Heart Disease (CHD) is one of the leading causes of death in the world. It happens when the coronary arteries that feed the heart muscle get narrowed or blocked by fatty deposits, a process called atherosclerosis. In the United States, the CDC reports heart disease as the number one cause of death, responsible for roughly 695,000 deaths every year (CDC, 2023). The World Health Organization puts the global figure at around 17.9 million cardiovascular deaths annually, and CHD accounts for the largest share of that number.

The tricky thing about CHD is that it builds up silently. For many patients, the first sign is a heart attack or severe chest pain, by which point the arteries are already badly blocked. If we can spot the people most at risk earlier, through lifestyle changes, medication, or timely procedures, a lot of those outcomes can be avoided. The catch is that the gold-standard tests for early CHD, such as coronary angiography or stress echocardiography, are expensive and usually need a specialist. In a primary care clinic, or in a rural setting, those options often are not available.

This is where machine learning (ML) fits in. Over the past few years, several studies have shown that ML models trained on ordinary clinical records can reach AUC-ROC values above 0.90 for heart disease prediction, which is good enough to be useful as a decision-support tool (Trigka & Dritsas, 2023; Dritsas & Trigka, 2023). The idea in this project is simple: can the kind of data that is already collected at a routine visit, things like age, blood pressure, cholesterol, and ECG readings, be used to flag high-risk patients without needing an angiogram first? The stakeholders who would benefit most are primary care physicians, cardiologists, and cardiothoracic surgeons, along with hospital teams that are trying to shorten the path from first visit to diagnosis.

1.2 Problem Statement

Even though the risk indicators for CHD are well known, a lot of patients are still diagnosed late. The information that could catch them earlier, blood pressure, cholesterol, maximum heart rate, resting ECG, is sitting in their records but is not being used in any systematic way at the primary care level. The more definitive tests, like coronary angiography, are expensive and often unavailable, so high-risk patients slip through at the first point of contact. This project looks at

whether a supervised ML model built on standard clinical features can close that gap and help clinicians prioritize who needs further cardiac testing.

1.3 Purpose of the Study

The goal of this study is to build and evaluate a supervised classification model that predicts whether a patient has CHD, using only the features available in a routine clinical record. The output is binary (disease or no disease), and the work is framed as a screening aid, not a diagnostic replacement. Alongside the prediction task, the study also examines which features drive the prediction, whether CHD-positive and CHD-negative patients differ on key clinical measures, and whether the model behaves fairly across male and female patients. The intent is to end up with something a clinician could actually use, or at least trust enough to consider, rather than a model that only looks good on paper.

2. Scope and Objectives

The study is built on the UCI Machine Learning Repository Heart Disease Dataset, which pools de-identified records from four centres: Cleveland (USA), Hungary, Switzerland, and VA Long Beach (USA). After cleaning, 918 patient records and 11 usable features remain. The target is binary, indicating whether a patient has heart disease (1) or not (0). The dataset is publicly available under a Creative Commons Attribution 4.0 licence, so no additional ethical approvals are needed for this work. All features are measurable clinical variables, which keeps the project grounded in things a clinician would actually record during a visit.

The ten-week capstone window is planned as follows: weeks 1 and 2 for data understanding and cleaning, weeks 3 and 4 for exploratory analysis, weeks 5 to 7 for model development and tuning, weeks 8 and 9 for evaluation and the fairness check, and week 10 for write-up and final submission. This timeline leaves a small buffer in case any step runs long, and keeps the scope realistic for a single-person project.

Four research questions drive the study. Each one is matched to a specific analytic technique in Section 5, so that every question is actually answerable with the data and the time available.

Research Question 1 (RQ1)

Which clinical features are the most statistically significant predictors of CHD, and how much do they contribute to model performance? This is answered with a mix of hypothesis tests (t-tests for

continuous variables, chi-square for categorical ones) and Random Forest feature importance scores. Effect sizes are reported along with p-values so that the answer is not just about statistical significance.

Research Question 2 (RQ2)

Among Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Support Vector Machine (SVM), and K-Nearest Neighbours (KNN), which algorithm classifies CHD best? Performance is measured with accuracy, precision, recall, F1, and AUC-ROC, with recall weighted most heavily because missing a real CHD case is worse than an unnecessary referral.

Research Question 3 (RQ3)

Do CHD-positive and CHD-negative patients differ significantly on resting blood pressure, cholesterol, and maximum heart rate? Independent-samples t-tests answer this, with Levene's test first for the equal-variance assumption, and Cohen's d for effect size.

Research Question 4 (RQ4)

Does gender change the relationship between clinical features and CHD, and does the imbalance in the data (around 79% male) hurt the model's fairness? This matters because the cardiology literature is clear that CHD risk factors and symptom presentation are different in women (Khoja et al., 2023). Separate confusion matrices and ROC curves are produced for male and female patients, and a chi-square test checks whether CHD prevalence itself differs by gender.

3. Sample Size Calculation

Sample size is worth taking seriously. If the study is under-powered, even a well-designed model can produce results that do not hold up (Button et al., 2013). The required sample size was calculated separately for each research question, using whichever method fit the question. The minimum sample size for the overall study is the largest of those four numbers.

A few assumptions are common across the four calculations and are worth justifying. The significance level is set at $\alpha = 0.05$, which is the standard Type I error rate in health and clinical research. Power is set at 0.80 (Cohen's convention), meaning the study accepts a 20% chance of missing a real effect in exchange for a manageable sample size. The minimum detectable effect for model comparison in RQ3 is set at a 5% absolute difference in accuracy, because a smaller difference than that is unlikely to matter clinically even if it is statistically detectable. The margin

of error for the confidence-interval methods is set at ±5%, which is tight enough to be useful and loose enough to be achievable with the available data.

3.1 Worked Example: RQ1

To make the approach concrete, here is the full calculation for RQ1, which uses the standard Wald confidence-interval formula for a single proportion. The proportion p = 0.553 is the observed prevalence of CHD in the cleaned dataset (508 positive cases out of 918). The formula is:

$$n = Z^2 \times p(1 - p) / e^2$$

Substituting Z = 1.96 (for α = 0.05, two-tailed), p = 0.553, and e = 0.05:

$$n = (1.96)^2 \times 0.553 \times (1 - 0.553) / (0.05)^2$$
$$n = 3.8416 \times 0.553 \times 0.447 / 0.0025$$
$$n = 0.9497 / 0.0025 = 379.9 \approx 380$$

So RQ1 needs at least 380 records to estimate the CHD prevalence within ±5% at 95% confidence. RQ4 uses the same confidence-interval formula with p = 0.85 (target recall for the female subgroup), which gives n = 196. RQ2 uses a two-proportion test for a small-to-medium effect size (Cohen's h = 0.30), which gives 88 per group. RQ3 is the strictest case: it is a two-proportion power test for detecting a 5% accuracy difference between two classifiers (p_1 = 0.85, p_2 = 0.80) at 80% power, which gives 906. The four results are summarized in Table 1.

Table 1. Sample Size Requirements by Research Question

RQ	Method	Key Parameters	Formula	Min N
RQ1	Confidence Interval	α = 0.05, Z = 1.96, p = 0.553, margin of error e = 0.05	$n = Z^2 \times p(1-p) / e^2$	**380**
RQ2	Two-Proportion Test	α = 0.05, Power = 0.80, Cohen's h = 0.30	n per group via power analysis	**88**
RQ3	Power-based (Accuracy)	α = 0.05, Power = 0.80, p_1 = 0.85, p_2 = 0.80 (5% accuracy difference)	Two-proportion power test	**906**
RQ4	Confidence Interval (Recall)	α = 0.05, Z = 1.96, p = 0.85, e = 0.05	$n = Z^2 \times p(1-p) / e^2$	**196**
Final	**Maximum across RQs**	The largest minimum sample size drives the required N for the study	$\max(N_i)$	**906**

Note. All power calculations assume two-tailed tests at α = 0.05 with 80% power where power-based methods are used.

The biggest number in the table is 906, which comes from RQ3 (model comparison at 80% power for a 5% accuracy difference). The cleaned dataset holds 918 records, so this threshold is cleared. With 5-fold stratified cross-validation, each fold uses about 734 records for training and 184 for testing, which is plenty for stable results. The Events-Per-Variable ratio is 50.8 (508 positive cases across 10 predictors), well above the usual minimum of 10 for logistic and tree-based models, so overfitting from too few events is not a concern here.

4. Data Description

4.1 Data Source

The dataset is the UCI Machine Learning Repository Heart Disease Dataset (Janosi et al., 1989; Dua & Graff, 2019), DOI 10.24432/C52P4X. It combines clinical records from four centres and is available under a CC BY 4.0 licence. No Kaggle data is used, in line with course policy.

Dataset URL: https://archive.ics.uci.edu/dataset/45/heart+disease

GitHub repository (code and cleaned data): https://github.com/mvinayabhushan/Capstone

Both the dataset and the accompanying code are publicly accessible to all stakeholders at the URLs above. No login, data-use agreement, or access request is required. Anyone involved in the review of this project can download the raw data from UCI and reproduce the cleaning and modelling steps from the GitHub repository.

4.2 Data Dictionary

Table 2 lists each feature retained in the cleaned dataset, its data type, and why it matters clinically. The target variable was built from the original multi-class severity rating by collapsing levels 1 through 4 into a single “Disease” class.

Table 2. Data Dictionary, UCI Heart Disease Dataset

Feature	Type	Description	Clinical Relevance
age	Continuous	Patient age in years	Older patients carry a higher baseline CHD risk

Feature	Type	Description	Clinical Relevance
sex	Binary	1 = Male, 0 = Female	Gender influences CHD prevalence and symptom presentation
cp	Categorical	Chest pain type (1–4)	Asymptomatic type (4) is strongly associated with CHD
trestbps	Continuous	Resting blood pressure (mmHg)	Elevated BP is a known cardiovascular risk factor
chol	Continuous	Serum cholesterol (mg/dl)	High cholesterol is linked to arterial narrowing and CHD
fbs	Binary	Fasting blood sugar > 120 mg/dl	Diabetes indicator; diabetes raises CHD risk
restecg	Categorical	Resting ECG results (0, 1, 2)	ST-T wave abnormalities indicate possible cardiac stress
thalach	Continuous	Maximum heart rate during exercise	A lower max HR during stress testing is associated with CHD
exang	Binary	Exercise-induced angina (1 = Yes)	Direct sign of reduced blood flow to the heart muscle
oldpeak	Continuous	ST depression (exercise vs. rest)	Higher values strongly indicate ischaemia and CHD risk
target_binary	Binary	0 = No Disease, 1 = Disease	Dependent variable used for classification

4.3 Data Cleaning Summary

The raw data had a few quality issues that needed sorting out before any modelling. Table 3 lists what was found and what was done about it. The approach follows standard data-cleaning practice (Rahm & Do, 2000; Bernardi et al., 2023).

Table 3. Data Cleaning Steps and Outcomes

Issue Identified	Action Taken	Outcome
2 duplicate records	Removed	918 clean records retained
Cholesterol = 0 (172 records)	Treated as missing, imputed with median (240)	No zero values remain

Issue Identified	Action Taken	Outcome
Resting BP = 0 (1 record)	Treated as missing, imputed with median (130)	No zero values remain
Missing values in trestbps, chol, fbs, restecg, thalach, exang, oldpeak (<10% each)	Imputed with column median	0 missing values in final dataset
slope (33.6% missing), ca (66.4% missing), thal (52.8% missing)	Columns dropped	11 features retained
Target had 5 severity levels (0–4)	Binarized: 0 = No Disease, 1 = Disease (levels 1–4 combined)	Balanced classes: 44.7% / 55.3%

5. Analytic Approach

The analysis uses a mix of inferential statistics and supervised machine learning. Where the question is about a difference between groups or an association between two variables, a statistical test is the right tool. Where the question is about prediction from a set of features, a classifier is. Each research question below states the hypothesis first, then the technique used to test it.

Before getting into each research question, the overall modelling pipeline is worth laying out in one place. The raw UCI data is first cleaned as described in Section 4.3. Preprocessing then splits by feature type: continuous variables (age, trestbps, chol, thalach, oldpeak) are standardized for the distance-based and margin-based models (KNN and SVM), since both are sensitive to feature scale, while tree-based models (Decision Tree, Random Forest, Gradient Boosting) and Logistic Regression use the raw values. Categorical variables (cp, restecg) are one-hot encoded. The cleaned and preprocessed dataset is then split 80/20 into training and test sets, with stratification on the target variable so that the disease prevalence is the same in both sets. Inside the training set, 5-fold stratified cross-validation is used for hyperparameter tuning with grid search. The final chosen model is then evaluated once on the held-out test set, which is never touched during training or tuning. This single-evaluation rule is important because it is the only unbiased estimate of real-world performance.

5.1 Solution to RQ1, Feature Significance

Hypothesis. CHD-positive patients will have lower maximum heart rates and higher ST depression than CHD-negative patients ($p < 0.05$), and both chest pain type and exercise-induced angina will be significantly associated with disease status.

Technique. Two angles, one statistical and one from the ML side. On the statistical side, independent-samples t-tests compare continuous features (age, trestbps, chol, thalach, oldpeak) between CHD-positive and CHD-negative groups, and chi-square tests of independence handle the categorical ones (sex, cp, fbs, restecg, exang). Cohen's d and Cramer's V are reported as effect sizes so that a tiny but statistically significant effect does not get mistaken for an important one. On the ML side, a Random Forest is trained on the full feature set, and its feature importance scores give a second opinion on which variables carry the most predictive weight. Pearson correlation is also computed for the continuous variables as a sanity check.

5.2 Solution to RQ2, Model Comparison

Hypothesis. Ensemble methods (Random Forest, Gradient Boosting) will beat single-learner baselines (Logistic Regression, Decision Tree, KNN) on both AUC-ROC and recall, which lines up with what the cardiovascular ML literature has found (Trigka & Dritsas, 2023).

Technique. Six classifiers are compared: Logistic Regression, Decision Tree, Random Forest (Breiman, 2001), Gradient Boosting (Friedman, 2001), SVM (Cortes & Vapnik, 1995), and KNN. These six were chosen on purpose to span the main families of supervised classification: linear (Logistic Regression), tree-based single-learner (Decision Tree), bagged ensemble (Random Forest), boosted ensemble (Gradient Boosting), margin-based (SVM), and instance-based (KNN). If one family is clearly better for this problem, the comparison will show it. Data is split 80/20 for training and testing, with 5-fold stratified cross-validation on the training split to check that performance is stable and not just a lucky split. Hyperparameters are tuned with grid search over a small documented grid (kept small on purpose, both for compute and to avoid overfitting to the validation folds). The final model is selected on AUC-ROC and recall on the held-out test set, with F1 as a tie-breaker.

5.3 Solution to RQ3, Group Differences

Hypothesis. Resting blood pressure, cholesterol, and maximum heart rate will all differ significantly between CHD-positive and CHD-negative groups ($p < 0.05$), and the largest effect size will show up for maximum heart rate.

Technique. Independent-samples t-tests compare the two groups on each of the three variables. Levene's test is run first to check whether the variances can be assumed equal; if not, Welch's t-test is used instead. Alpha is set at 0.05. Cohen's d is reported for any significant difference so that the size of the effect is visible, not just its existence.

5.4 Solution to RQ4, Gender Moderation

Hypothesis. Recall will be lower for female patients than for male patients because the training data is about 79% male. This is consistent with what has been documented in cardiology: CHD in women is often under-recognised because the risk profiles and symptom patterns are not the same as in men (Khoja et al., 2023).

Technique. Separate confusion matrices and AUC-ROC curves are generated for male and female patients on the test set. A chi-square test of independence checks whether CHD prevalence itself differs significantly by gender in the data. If a subgroup performance gap shows up, it is documented as an explicit fairness limitation in the recommendations rather than glossed over. The documentation approach follows the datasheets-for-datasets framework from Gebru et al. (2021), which argues for making dataset-level limitations visible as part of the deliverable, not footnotes.

5.5 Evaluation Metrics

Table 4 lists every metric that will be computed for each model, along with the formula. In a clinical screening context, recall (sensitivity) is the one to watch. Missing a real CHD case is a much costlier error than referring a healthy patient for an extra test, so the threshold and model selection both favour recall.

Table 4. Model Evaluation Metrics with Formulae

Metric	Formula	Why It Matters
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Overall proportion of correct predictions
Precision	TP / (TP + FP)	Of all predicted positives, how many are real CHD cases?
Recall (Sensitivity)	TP / (TP + FN)	Of all actual CHD cases, how many did the model catch?

Metric	Formula	Why It Matters
Specificity	TN / (TN + FP)	Of all actual non-CHD cases, how many were correctly ruled out?
F1 Score	2 × (Precision × Recall) / (Precision + Recall)	Balances precision and recall in a single value
AUC-ROC	Area under the ROC curve	Overall ability to separate CHD from non-CHD across thresholds

Abbreviations: TP = True Positive, TN = True Negative, FP = False Positive, FN = False Negative. The classification threshold is not left at the default 0.5. Instead, it is tuned on the validation split to maximize recall subject to a minimum precision floor, so that false negatives are kept low without flooding clinicians with low-value referrals.

6. Recommendations and Application

6.1 Target Users and Intended Use

The model is meant to be a clinical decision-support tool, not a replacement for a clinician's judgement. Two user groups are in mind from the start.

Primary care physicians see most patients first. For them, the tool acts as an early screen. Once the patient's routine readings are in (age, blood pressure, cholesterol, ECG, maximum heart rate), the model returns a CHD risk score. A high score is a prompt to fast-track that patient for further cardiac work-up, such as an echocardiogram or a cardiology referral, instead of waiting for classic symptoms to show up. That matters most in rural and resource-constrained settings, where specialist access can take weeks.

Cardiologists and cardiothoracic surgeons use the tool in a different way. For a cardiologist running a busy outpatient list, the risk score helps with triage and with justifying invasive tests like angiography for the patients most likely to benefit. For a cardiothoracic surgeon, or any surgeon really, it is useful before non-cardiac surgery, where undetected CHD is a known driver of perioperative complications. In either case, the model is best thought of as a second opinion built on patterns from hundreds of past cases, which can either back up the clinician's first impression or push them to look again.

6.2 Deployment Considerations

For a hospital or health system, the cleanest way to deploy this is as a module inside the electronic health record (EHR). The risk score is computed in the background whenever the required fields are filled in, so it shows up as part of the normal workflow rather than as a separate tool the clinician has to remember to open. Three practical constraints shape that deployment. First, data privacy. Patient data used for scoring has to stay inside the hospital's protected environment, and the model's predictions need an audit trail, to meet HIPAA in the US, GDPR in Europe, and the equivalents elsewhere. Second, regulatory approval. Before being used for formal clinical decisions (as opposed to research or internal quality improvement), the tool would need the right regulatory clearance, such as the FDA's Software-as-a-Medical-Device pathway in the US. Third, clinician trust. A model that scores well on paper but that clinicians do not understand or trust will simply not be used, and that is a people-and-training problem as much as a technical one.

6.3 Model Limitations and Clinician Interpretation

A few limitations are worth being upfront about. The training data is about 79% male, so recall for female patients is likely to be lower than for male patients. RQ4 is designed precisely to measure that gap so it cannot be hidden. The dataset also comes from four specific centres and from a specific time period, which means the model should be revalidated on local data before being relied on in a new setting. And the feature set is only what the UCI dataset contains: newer biomarkers like high-sensitivity troponin or coronary calcium scores are not in there, and they would probably add predictive value if they were.

For clinicians using the output, the risk score is one input, not the answer. A high-confidence positive score is a reason to investigate further, but a low or borderline score is not a reason to rule out CHD in a patient who is symptomatic. Since the threshold is tuned to minimize false negatives, there will be some false positives; that is the trade-off the design is deliberately making. Calibration plots will be shipped with the model so a clinician can see, for each probability level, what fraction of those patients actually had CHD in the validation data. Over time, if the tool is deployed, outcomes for flagged patients should be logged and fed back in, so the model can be monitored for drift and retrained when needed.

7. References and Bibliography

Bernardi, F. A., Alves, D., Crepaldi, N., Yamada, D. B., Lima, V. C., & Rijo, R. (2023). Data quality in health research: Integrative literature review. Journal of Medical Internet Research, 25, e41446. https://doi.org/10.2196/41446

Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5–32. https://doi.org/10.1023/A:1010933404324

Button, K. S., Ioannidis, J. P. A., Mokrysz, C., Nosek, B. A., Flint, J., Robinson, E. S. J., & Munafò, M. R. (2013). Power failure: Why small sample size undermines the reliability of neuroscience. Nature Reviews Neuroscience, 14(5), 365–376. https://doi.org/10.1038/nrn3475

Centers for Disease Control and Prevention. (2023). Coronary artery disease (CAD). U.S. Department of Health and Human Services. https://www.cdc.gov/heartdisease/coronary_ad.htm

Cortes, C., & Vapnik, V. (1995). Support-vector networks. Machine Learning, 20(3), 273–297. https://doi.org/10.1007/BF00994018

Dritsas, E., & Trigka, M. (2023). Efficient data-driven machine learning models for cardiovascular diseases risk prediction. Sensors, 23(3), 1161. https://doi.org/10.3390/s23031161

Dua, D., & Graff, C. (2019). UCI Machine Learning Repository: Heart Disease Dataset [Data set]. University of California, Irvine. https://archive.ics.uci.edu/dataset/45/heart+disease

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. Annals of Statistics, 29(5), 1189–1232. https://doi.org/10.1214/aos/1013203451

Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Daumé, H., & Crawford, K. (2021). Datasheets for datasets. Communications of the ACM, 64(12), 86–92. https://doi.org/10.1145/3458723

Janosi, A., Steinbrunn, W., Pfisterer, M., & Detrano, R. (1989). International application of a new probability algorithm for the diagnosis of coronary artery disease. American Journal of Cardiology, 64(5), 304–310. https://doi.org/10.1016/0002-9149(89)90524-9

Khoja, A., Andraweera, P. H., Lassi, Z. S., Ali, A., Zheng, M., Pathirana, M. M., Aldridge, E., Wittwer, M. R., Chaudhuri, D. D., Tavella, R., & Arstall, M. A. (2023). Risk factors for premature coronary heart disease in women compared to men: Systematic review and meta-analysis. Journal of Women's Health, 32(9), 908–920. https://doi.org/10.1089/jwh.2022.0517

Rahm, E., & Do, H. H. (2000). Data cleaning: Problems and current approaches. IEEE Data Engineering Bulletin, 23(4), 3–13.

Trigka, M., & Dritsas, E. (2023). Long-term coronary artery disease risk prediction with machine learning models. Sensors, 23(3), 1193. https://doi.org/10.3390/s23031193